SERVER

```
import socket
s=socket.socket()
s.bind(('127.0.0.1',12345))
s.listen(1)
Print("Server is Listnening...")
c,addr=s.accept()
print(c.recv(1024).decode())
c.send("Hello Client".encode())
print(c.recv(1024).decode())
c.close()
```

CLIENT

```
import socket
s=socket.socket()
s.connect(('127.0.0.1',12345))
s.send("Hello Server".encode())
print(s.recv(1024).decode())
s.send("Thank you Server".encode())
s.close()
```

CAESAR

```
text="ATTACKATONCE"
k=4
c=""
for ch in text:
    c+=chr((ord(ch)-65+k)%26+65)
print("Cipher:",c)
p=""
for ch in c:
    p+=chr((ord(ch)-65-k)%26+65)
print("Plain:",p)
```

HILL CIPHER

```
key="GYBNQKURP"
msg="ACT"
def n(c): return ord(c)-65
K=[[n(key[i*3+j]) for j in range(3)] for i in range(3)]
M=[n(c) for c in msg]
for i in range(3):
    x=(K[i][0]*M[0]+K[i][1]*M[1]+K[i][2]*M[2])%26
    print(chr(x+65),end="")
```

PLAYFAIR CIPHER

```
key="MONARCHY"
text="INSTRUMENTS"
alpha="ABCDEFGHIKLMNOPQRSTUVWXYZ"
table=""
for c in key+alpha:
    if c not in table and c!="J":
        table+=c
pairs=[]; i=0
while i<len(text):
    a=text[i]
    b=text[i+1] if i+1<len(text) else "X"
    if a==b: pairs.append(a+"X"); i+=1
    else: pairs.append(a+b); i+=2
cipher=""
for p in pairs:
    r1,c1=divmod(table.index(p[0]),5)
    r2,c2=divmod(table.index(p[1]),5)
    if r1==r2:
cipher+=table[r1*5+(c1+1)%5]+table[r2*5+(c2+1)%5]
    elif c1==c2:
cipher+=table[((r1+1)%5)*5+c1]+table[((r2+1)%5)*5+c
2]
    else:
        cipher+=table[r1*5+c2]+table[r2*5+c1]
print(cipher)
```

DES

```
from Crypto.Cipher import DES
from Crypto.Util.Padding import pad,unpad

key=b'12345678'
data=b'HELLODES'

cipher=DES.new(key,DES.MODE_ECB)
enc=cipher.encrypt(pad(data,8))
print("Encrypted:",enc)

dec=unpad(cipher.decrypt(enc),8)
print("Decrypted:",dec.decode())
```

AES

```
from Crypto.Cipher import AES
from Crypto.Util.Padding import pad,unpad

key=b'Sixteen byte key'
data=b'HELLOAES'

cipher=AES.new(key,AES.MODE_ECB)
enc=cipher.encrypt(pad(data,16))
print("Encrypted:",enc)

dec=unpad(cipher.decrypt(enc),16)
print("Decrypted:",dec.decode())import socket
```

SERVER

```
s=socket.socket()
s.bind(("localhost",1234))
s.listen(1)
print("Waiting...")
c,a=s.accept()
print(c.recv(1024).decode())
c.close()
```

CLIENT

```
import socket
s=socket.socket()
s.connect(("localhost",1234))
s.send("Hello Packet".encode())
s.close()
```